Decoding of handwritten English sentences is rational

Background. Handwriting remains a critical medium of communication, from education and medicine to administration and everyday transactions. How do readers decipher such input? We hypothesize that they rationally integrate bottom-up evidence with top-down expectations about likely meanings, following the "noisy channel processing" framework [1-3]. Such integration is rational when the two information sources trade off: lower input quality (higher uncertainty), or increased likelihood of "corruptions", should lead comprehenders to weigh prior knowledge more heavily (e.g., [1, 4-6]). To stress-test this idea, we exposed readers to a wide variety of handwriting styles, forcing them to repeatedly re-balance this trade-off from minimal evidence.

Methods. We tested how readers interpreted handwritten sentences that were implausible but could become plausible by changing one letter (e.g., "*mom got a dog but the baby was upset by all the banks*", which could be "*barks*"). We manipulated input uncertainty in two ways via a mixed 2×2 design: legibility (high or low; between subjects) and the visual similarity between the observed letter and its plausible alternative (dissimilar like g-m, or similar like n-r, based on [7]; within subjects) (**Figure 1**). To this end, we created 20 sentences. The implausible letter did not occur anywhere in the sentence other than in the critical word, and the alternative letter did not occur at all, to prevent participants from using other occurrences as cues to visual decoding. The word with the implausible letter was higher in frequency than its plausible alternative, so any non-literal interpretations (e.g., decoding "*banks*" as "*barks*") could not reflect a frequency bias.

Before the main experiment, each sentence (and each of 60 plausible fillers) was handwritten by 28 undergraduate students (n=56 total). For each critical sentence, we chose one legible and one "illegible" (lower legibility) image. In a norming phase, 95 new students were asked to rate image legibility and transcribe the text literally (by typing). For each critical item, half the participants saw the full sentence, and half—only the critical word (out of context), to ensure that the bottom-up visual input was perceived as the intended word ("*banks*") and not its plausible alternative ("*barks*"). In the main experiment, 343 participants on Prolific were asked to transcribe the images. Transcriptions were coded as "literal" (typing the sentence with the implausible word) or "non-literal" (typing the plausible alternative) (other responses were excluded) (**Figure 2**).

Results. For the norming data (**Figure 3**), we modeled legibility ratings with a mixed-effects, linear regression (**Equation 1**; the maximal model that converged). The legible materials were rated as more legible than their illegible counterparts, when presented as either sentences (b=1.25, SE=0.06, Z=20.37, p<0.001) or isolated words (b=0.87, SE=0.07, Z=13.3, p<0.001).

For the main experiment, we modeled participants' responses using a mixed-effects logistic regression (**Equation 2**; the maximal model that converged). Consistent with our predictions, we found two effects (**Figure 2**): first, lower legibility increased non-literal transcriptions (b=2.04, SE=0.38, Z=5.42, p<0.001). Second, higher visual similarity between the implausible letter and its plausible alternative also increased non-literal transcriptions (b=2.82, SE=0.52, Z=5.41, p<0.001). These effects did not interact (b=0.24, SE=0.29, Z=0.82).

Discussion. Participants rationally deciphered handwriting, relying more on their top-down expectations for plausible meanings when input quality was compromised (low legibility) or corruptions were perceived as more likely (high letter similarity). Our study goes beyond past work because the ambiguity of handwriting exhibited high variability across trials (writers), in contrast to typed text in past studies where the kinds of noise are highly constrained [2]. Even studies using spoken stimuli usually introduce a constant level of "noise" per participant [4]. Moreover, even though auditory speech in the real world (cf. in experiments) does exhibit high variability across speakers, handwriting may be more challenging still: on average, hearing and sighted individuals have much less experience comprehending handwriting vs. speech, and the number of writers they are exposed to is much smaller than the number of speakers. Therefore, our results provide a stringent test of noisy-channel comprehension.

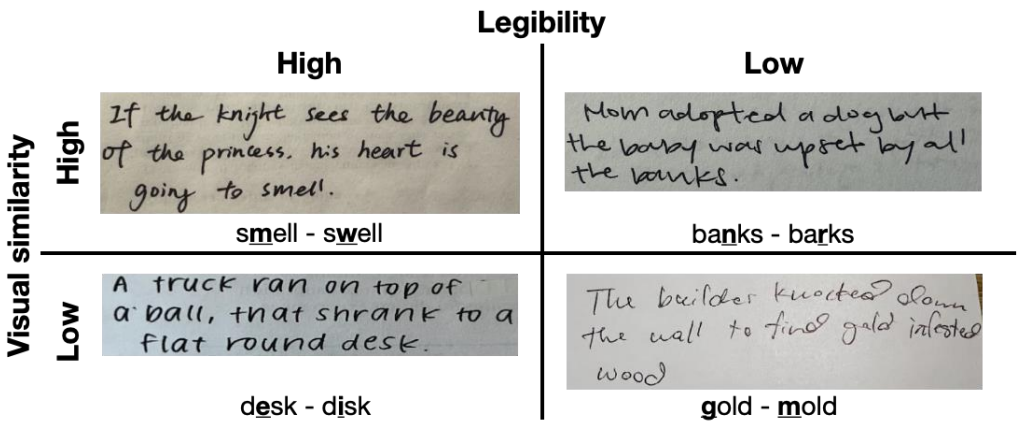

Figure 1. Sample stimuli.

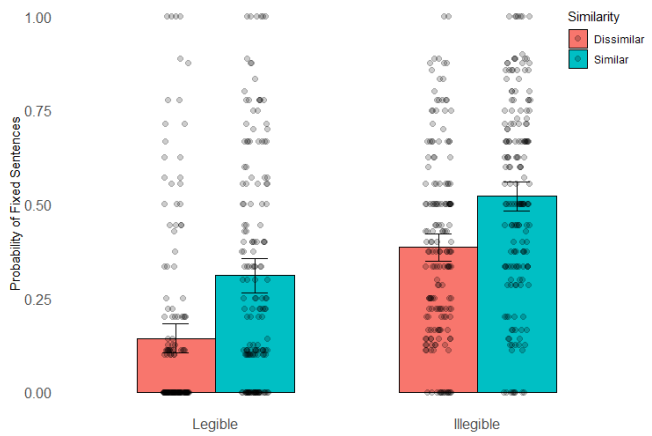

Figure 2. Main results. The *y*-axis shows the probability that a participant would "fix"" a sentence by transcribing it non-literally. The legibility of the handwriting is shown on the *x*-axis, and letter similarity is shown in color. Bars show averages, error bars show standard errors of the mean. Dots represent individual participant.

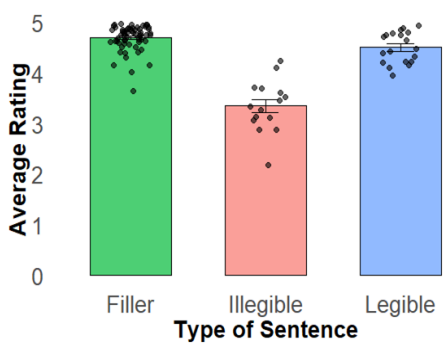

Figure 3. Norming results for full sentences. Dots represent individual items. Other conventions are the same as in Figure 2.

Equation 1. Rating ~ 1 + legibility × presentation + (1 + legibility × presentation | participant) + (1 + presentation | item)

Legibility is binary (high vs. low), presentation is binary (sentence vs. isolated word). The symbol × denotes both the two fixed effects and the interaction.

Equation 2. IsLiteral ~ 1 + legibility × letterSimilarity + (1 | participant) + (1 + legibility | item)

Legibility and letterSimilarity are binary.

References. [1] Levy *et al.* (2009), *PNAS*; [2] Gibson *et al.* (2013), *PNAS*; [3] Ding *et al.* (2025), *EMNLP*; [4] Yurovsky *et al.* (2017), *Psych. Sci.*; [5] Gibson *et al.* (2017), *Psych. Sci.*; [6] Ryskin *et al.* (2025), *Psych. Bull. Rev.*; [7] Simpson *et al.* (2012), *Beh. Res. Methods*.